ASSIGNMENT 1

1. **BAR CHART - Demand comparison chart**

   Purpose: A bar chart is used to represent data visually using rectangular bars. It helps compare values across different categories easily.

   STEPS: Open Tableau Public, Load dataset. Click Sheet 1.

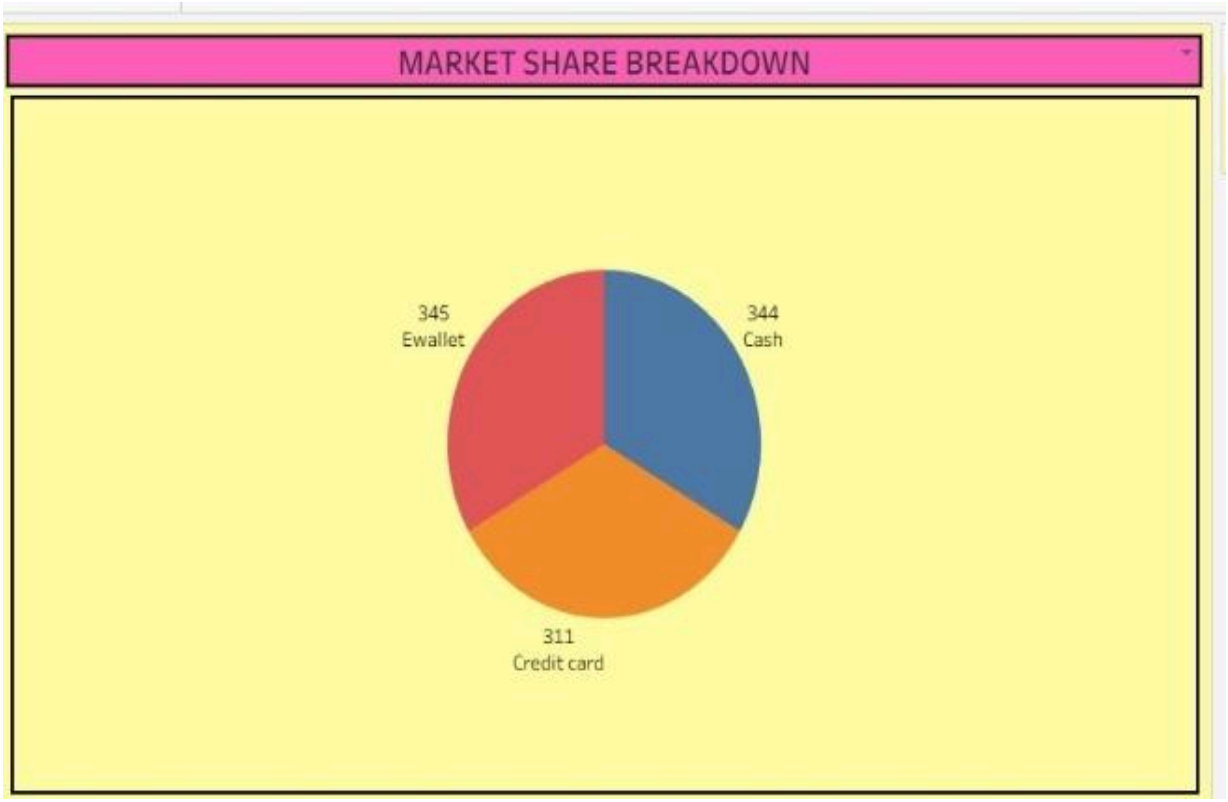
   - Next, Drag Product line → to Columns and Drag Total (Sales/Amount) → to Rows (vertical bars will automatically appear)
   - In the Marks card , select Bar.
   - On the Marks card → Click Label. Tick Show Mark Labels. Now numbers (56,145 etc.) appear on top of bars.
   - Sort Bars (Descending) using sort icon.
   - Formatting : Change Bar Color Click Color (Marks card) Choose Blue .
   - Change Background Color, Go to Format → Shading.
   - Under Worksheet → Pane → Select light green (like screenshot).
   - Format Title, Double-click title give title as Demand comparison chart Click OK.
   - To change Color-Go to Format → Font → Title → Choose Color (purple/blue).
   - Go to Format →Give Borders.

2. PIE CHART - market share breakdown

Purpose: A pie chart is used to represent data as parts of a whole. It shows how each category contributes to the total using slices of a circle.

STEPS: Open Tableau Public, Load dataset. Click Sheet 2.

- Go to Marks Card, Select Pie.
- Drag Payment to the colour in the mark Card.
- Drag Payment Again to Label, right click label and Tick Show mark Label.
- Drag CNT(payment) To angle and Label.
- For percentage, go Show Percentage on Total on label.
- Change Background Color, Go to Format → Shading.
- Under Worksheet → Pane → Select light green (like screenshot).
- Format Title, Double-click title give title as market share breakdown Click OK.
- To change Color-Go to Format → Font → Title → Choose Color (purple/blue).
- Go to Format →Give Borders



3. **STACKED BAR CHART - proportional contribution bars**

Purpose: A stacked bar chart is used in branch-wise sales to compare total sales of each branch and also see the contribution of different products or categories within that branch.

STEPS: Open Tableau Public, Load dataset. Click Sheet 3.

- Drag Branch to Columns And Drag Total (or Sales/Amount) to Rows .
- Now see simple vertical bars for each branch.
- Go to Marks card (left side), Click dropdown (Automatic) Select Bar.
- Drag Gender to Color (in Marks card).(Now bars split into Male & Female colors.)
- Drag Gender to Label (Marks card). Then Right Click Label Show mark labels.
- Go to Format → Shading. Under Worksheet → Pane → Select Colour
- Format Title, Double-click title give title as proportional contribution bars and Click OK.
- Go to Format → Font → Title → Choose Colour ,Then Go to border and give border.

4. **LINE CHART - performance Trend line**

Purpose: A line chart for monthly data is used to show how values change from month to month across a year or any time period.

STEPS: Open Tableau Public, Load dataset. Click Sheet 4.

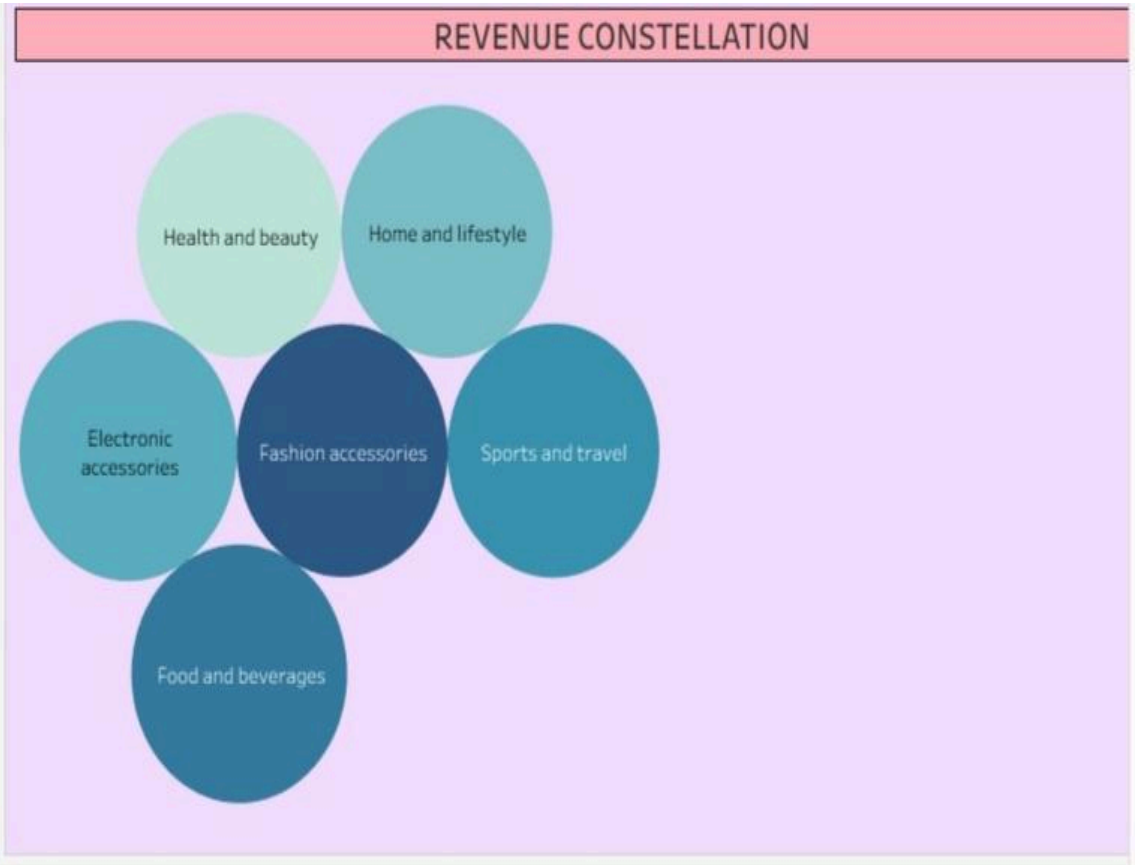
- Drag Date to Columns and Select MONTH(Date).
- Drag Sum Total to Rows.
- Go to Marks Select Line Then Drag SUM(Total) to Label and Tick Show Mark Labels, Automatically Values appear on line.
- Format Line Color Choose single Color and Size Increase thickness.
- Go to Format → Shading. Under Worksheet → Pane → Select Colour
- Format Title, Double-click title give title as performance Trend line and Click OK.
- Go to Format → Font → Title → Choose Colour ,Then Go to borders and give borders

5. **BUBBLE CHART - revenue constellation**

Purpose: A Bubble Chart is used to visualize and compare product line performance using circles (bubbles).Each bubble represents one product category.

Steps: Open Tableau Public, Load dataset. Click Sheet 5.

- Go to the Marks card and change from Automatic Circle.
- This converts the view into a bubble-style chart or select bubble chart from Show me.
- Drag Product line Label And Drag SUM(Quantity) to Size.
- Increase the Size slider to make bubbles more visible.
- Click Label and Tick Show Mark Labels.
- Format Axes for Right-click Unit price axis → Edit Axis and Give Title Unit Price
- Drag SUM(unit price) to colour
- Go to Format → Shading. Under Worksheet → Pane → Select Colour .
- Format Title, Double-click title give title as revenue constellation and Click OK.
- Go to Format → Font → Title → Choose Colour ,Then Go to borders and give borders



ASSIGNMENT 2

1. **DONUT CHART: CUSTOMER SEGMENT DISTRIBUTION**
   Purpose:--To represent part-to-whole relationships by showing how each category contributes to the total value.

   Steps:-- Open Tableau Public, Click Connect → Text file / Excel, Load supermarket_sales dataset Go to Sheet 1

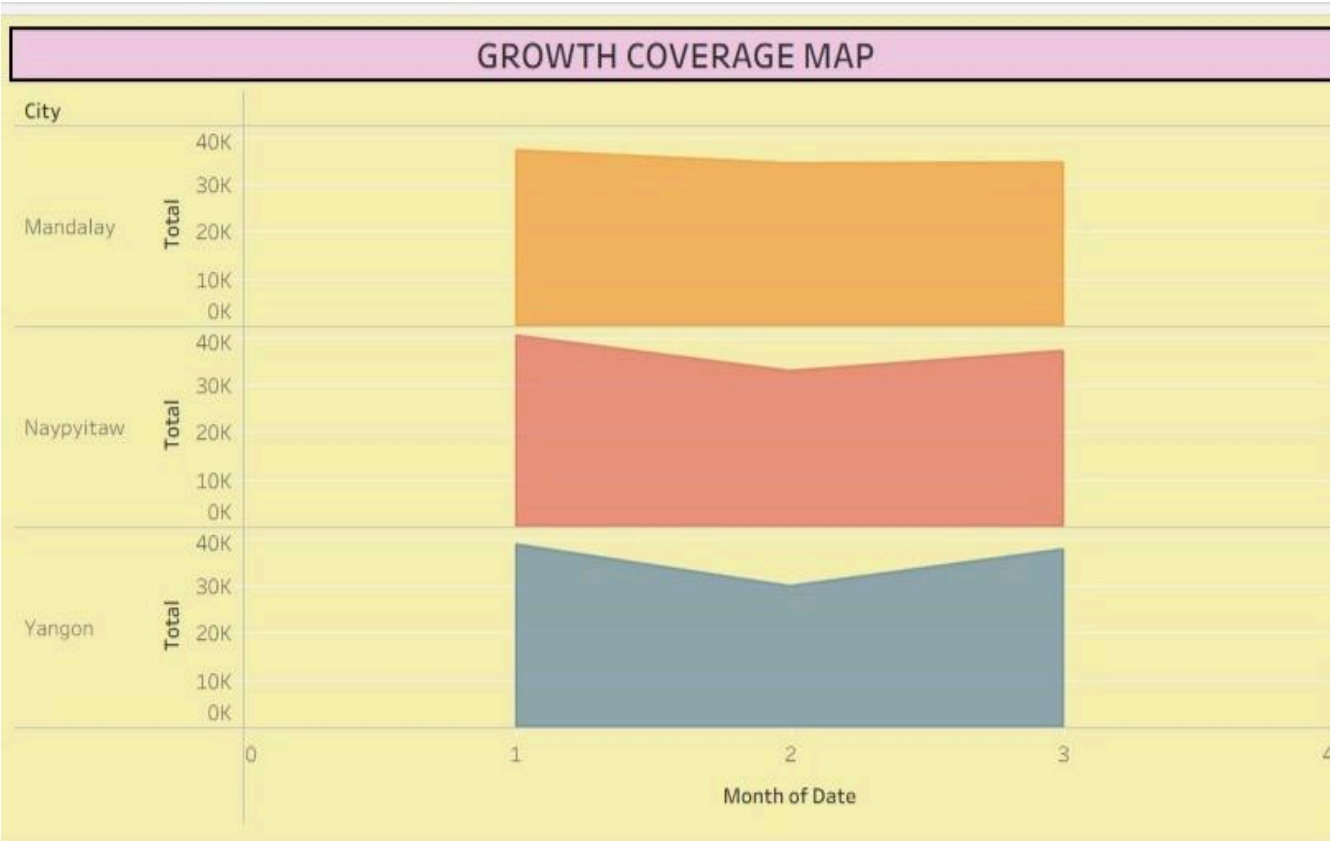
   - ➢ Drag Category to Color (Marks card) then Drag Total to Angle and Drag Total → Label
   - ➢ Change Marks type → Pie and For Create Dual Pie , Drag Number of Records → Rows (twice) Right-click second axis → Dual Axis
   - ➢ Click 2nd Marks card (AVG(Number of Records)) and Remove Category from Color
   - ➢ Change Color to White (or background Color) and Reduce Size Make it smaller
   - ➢ Go to 2nd Marks card, Drag Number of Records to Label
   - ➢ Edit label to Increase font size and Align Center
   - ➢ Go to Format → Shading. Under Worksheet → Pane → Select Colour .
   - ➢ Format Title, Double-click title give title as customer segment distribution and Click OK.
   - ➢ Go to Format → Font → Title → Choose Colour ,Then Go to borders and give borders

2. AREA CHART: GROWTH COVERAGE MAP

Purpose: To show how different categories contribute to the total over time and compare their trends and growth.

STEPS: Open Tableau Public, Click Connect → Text file / Excel, Load supermarket_sales dataset Go to Sheet 2

- Drag Date → Columns → Change to MONTH(Date)
- Drag Total → Rows
- Drag City → Rows (place before SUM(Total))
- Change Marks type → Area
- Drag Branch → Color (Marks card)
- Adjust Size if needed from Size option
- Go to Format → Shading. Under Worksheet → Pane → Select Colour .
- Format Title, Double-click title give title as growth coverage map and Click OK.
- Go to Format → Font → Title → Choose Colour ,Then Go to borders and give borders

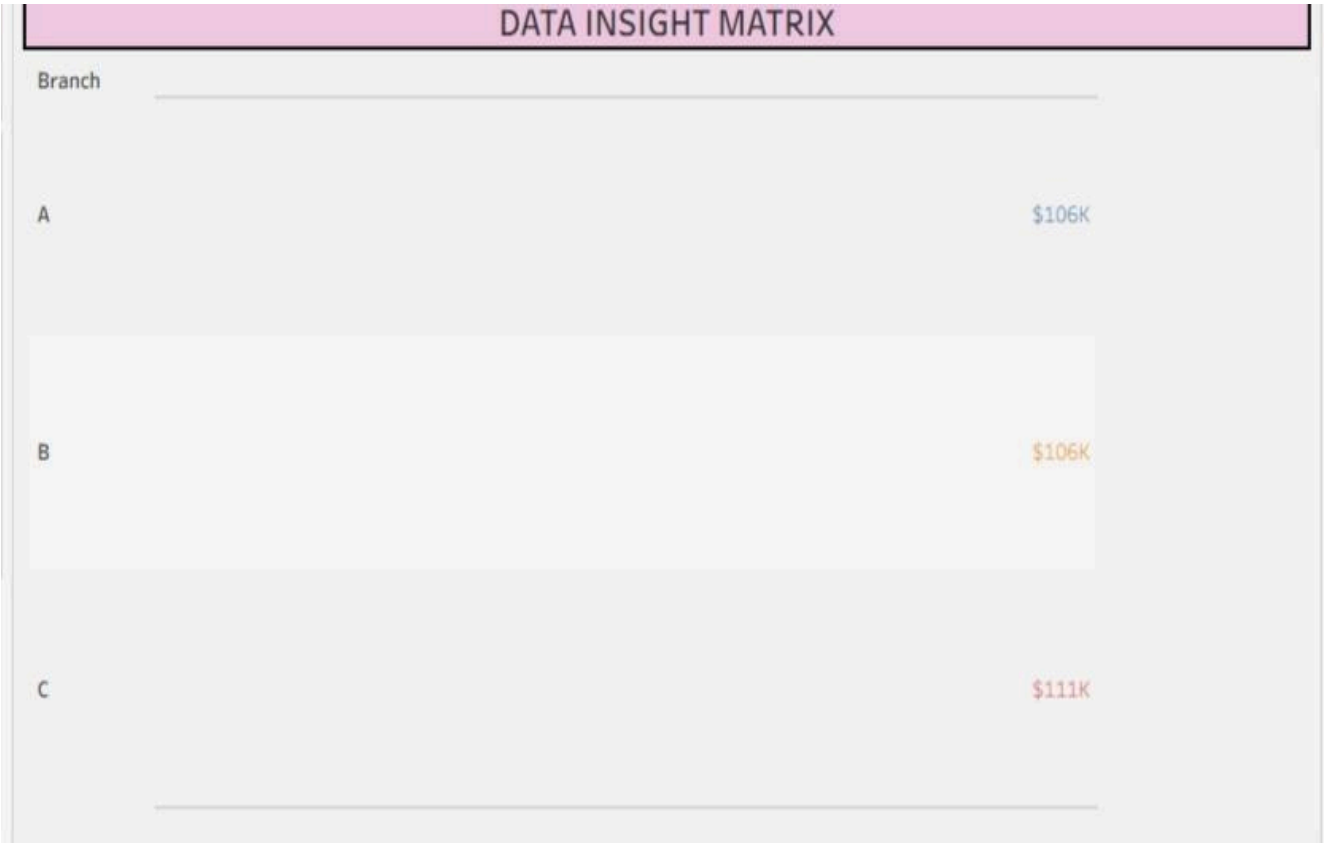


3. TEXT TABLE: DATA INSIGHT MATRIX

Purpose:--To display exact data values in a tabular format for detailed comparison and reporting.

Steps:--Open Tableau Public, Click Connect → Text file / Excel, Load supermarket_sales dataset Go to Sheet 3.

- ➢ From Data pane Drag Branch to Rows Then Drag Total →to Text (Marks card)
- ➢ Go to Marks dropdown Select Text
- ➢ Right-click SUM(Total) Click Format , Under Numbers Choose Currency (Custom) , Display Units Thousands (K)
- ➢ Go to Format → Shading. Under Worksheet → Pane → Select Colour .
- ➢ Format Title, Double-click title give title as data insight matrix and Click OK.
- ➢ Go to Format → Font → Title → Choose Colour ,Then Go to borders and give borders



4. HIGHLIGHTED TABLE: KPI HIGHLIGHT DASHBOARD

Purpose:-- To compare sales performance across product segments using color intensity to identify high and low values quickly.

Steps:--Open Tableau Public → Click Connect → Text file / Excel → Load supermarket_sales dataset and Go to Sheet 4.

- ➢ Drag Product line to Columns Then Drag Branch to Rows.
- ➢ Drag Total to Color (Marks card) Next Drag Total Label
- ➢ Change Marks type Square (Or click Show Me → Highlight Table)
- ➢ Click Label Show mark labels Format numbers to Currency
- ➢ Go to Format → Shading. Under Worksheet → Pane → Select Colour .
- ➢ Format Title, Double-click title give title as kpi highlight dashboard and Click OK.
- ➢ Go to Format → Font → Title → Choose Colour ,Then Go to borders and give borders

KPI HIGHLIGHT DASHBOARD

	Product line					
Branch	Electronic accessories	Fashion accessories	Food and beverages	Health and beauty	Home and lifestyle	Sports and travel
A	18,317	16,333	17,163	12,598	22,417	19,373
B	17,051	16,413	15,215	19,981	17,549	19,988
C	18,969	21,560	23,767	16,615	13,896	15,762

5. WORD CLOUD: DATA EXPLANATION CLOUD

Purpose:-- To visualize text data by displaying frequently occurring words in larger and more prominent sizes.

Steps:--Open Tableau Public → Click Connect → Text file / Excel → Load supermarket_sales dataset and Go to Sheet 5.

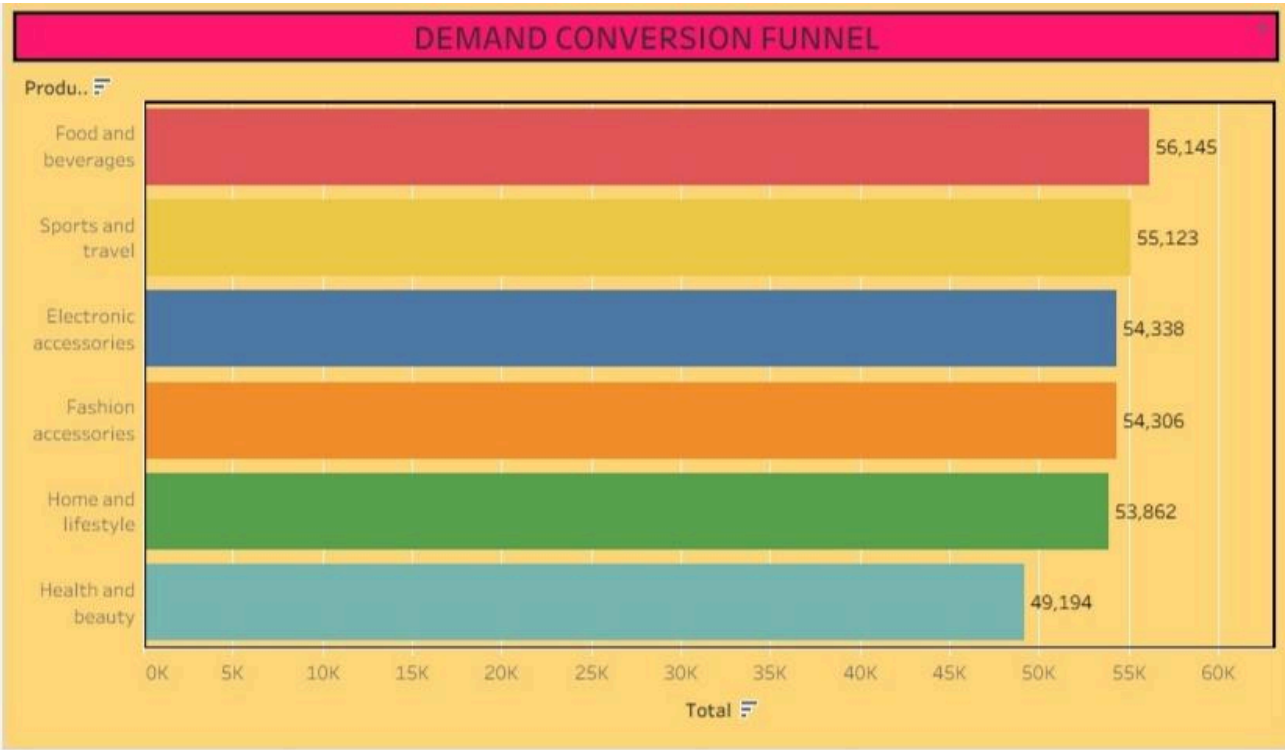
- Go to Marks card, Select Text.
- Drag Product line to Text (Marks card) Now product names appear as words.
- Drag Quantity (or Sales / Total) to Size.
- Drag Product line to Color(Now each word has a different color).
- Click Size , Move slider → Increase for big bold words.
- Go to Format → Shading. Under Worksheet → Pane → Select Colour .
- Format Title, Double-click title give title as data expression cloud and Click OK.
- Go to Format → Font → Title → Choose Colour
- Then Go to borders and give borders

6. FUNNEL CHART: DEMAND CONVERSATION FUNNEL

Purpose:-- A Funnel Chart is used to show how data moves through different stages and how the value decreases step-by-step.

STEPS:-- Open Tableau Public → Click Connect → Text file / Excel → Load supermarket_sales dataset and Go to Sheet 6.

- ➢ Drag Product line to Rows. Then Drag SUM(Total) Columns. Now Horizontal bars appear.
- ➢ Click Sort Descending icon
- ➢ Now the biggest bar on top (funnel shape order).
- ➢ Go to the Marks card, Select Bar .
- ➢ Drag Product line to Color. Now each bar gets a different color.
- ➢ Click Size, Drag slider for Increase width.
- ➢ Go to Format → Shading. Under Worksheet → Pane → Select Colour .
- ➢ Format Title, Double-click title give title as demand conversation funnel and Click OK.
- ➢ Go to Format → Font → Title → Choose Colour
- ➢ Then Go to borders and give borders

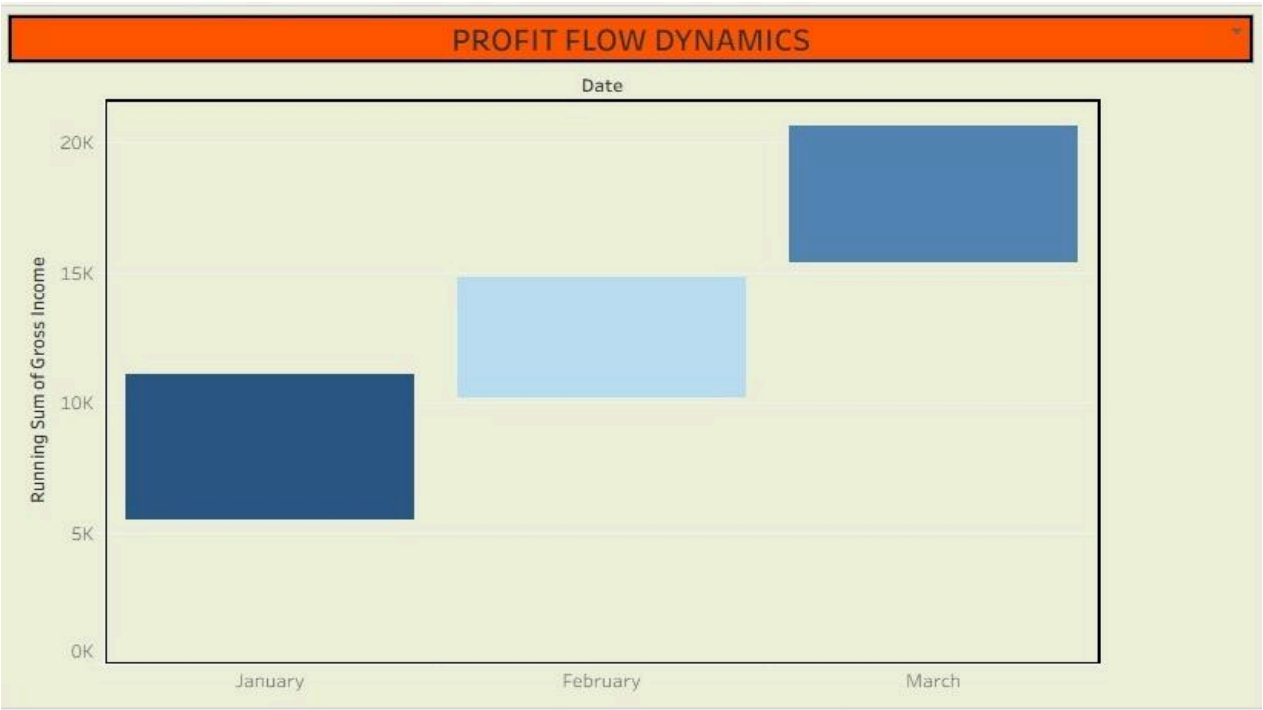


7. WATERFALLS: PROFIT FLOW DYNAMICS

Purpose:-- A Waterfall Chart shows how an initial value changes step-by-step due to increases and decreases. It explains how the final value is reached.

Steps:--Open Tableau Public → Click Connect → Text file / Excel → Load supermarket_sales dataset → Go to Sheet 7.

- ➢ Drag Date to Columns and Change to MONTH(Date).
- ➢ Drag Gross Income to Rows.
- ➢ Right-click SUM(Gross Income) , Select Quick Table Calculation → Running Total.
- ➢ Go to Marks card → Change Marks type → Gantt Bar.
- ➢ Drag SUM(Gross Income) to SizeDrag SUM(Gross Income) to Color.
- ➢ Go to Format → Shading. Under Worksheet → Pane → Select Colour .
- ➢ Format Title, Double-click title give title as profit flow dynamics and Click OK.
- ➢ Go to Format → Font → Title → Choose Colour .
- ➢ Then go to borders and give borders .



ASSIGNMENT-2

Long Term Virtual Internship Data Analytics using Tableau

1. Prepare Donut Chart:
PURPOSE:

- Understand total sales performance
- Compare sales across product categories

STEPS:

Step 1: Connect Data

1. Open Tableau
2. Click Connect → Select your dataset (Excel/CSV, etc.)

Step 2: Create Basic Pie Chart

1. Drag Category → Rows
2. Drag Sales → Marks → Angle
3. Change Marks type to Pie
4. Drag Sales → Label (to show values)
5. Drag Category → Color

Step 3: Convert Pie to Donut Chart

1. Create a calculated field:
   ZERO = 0
2. Drag ZERO to Rows twice
3. Right-click second ZERO → Dual Axis
4. On the second Marks card (inner circle):
   - Change to Pie
   - Remove Category
   - Make it White color
   - Reduce size (this creates the hole effect)

Step 4: Show Total Sales in Center

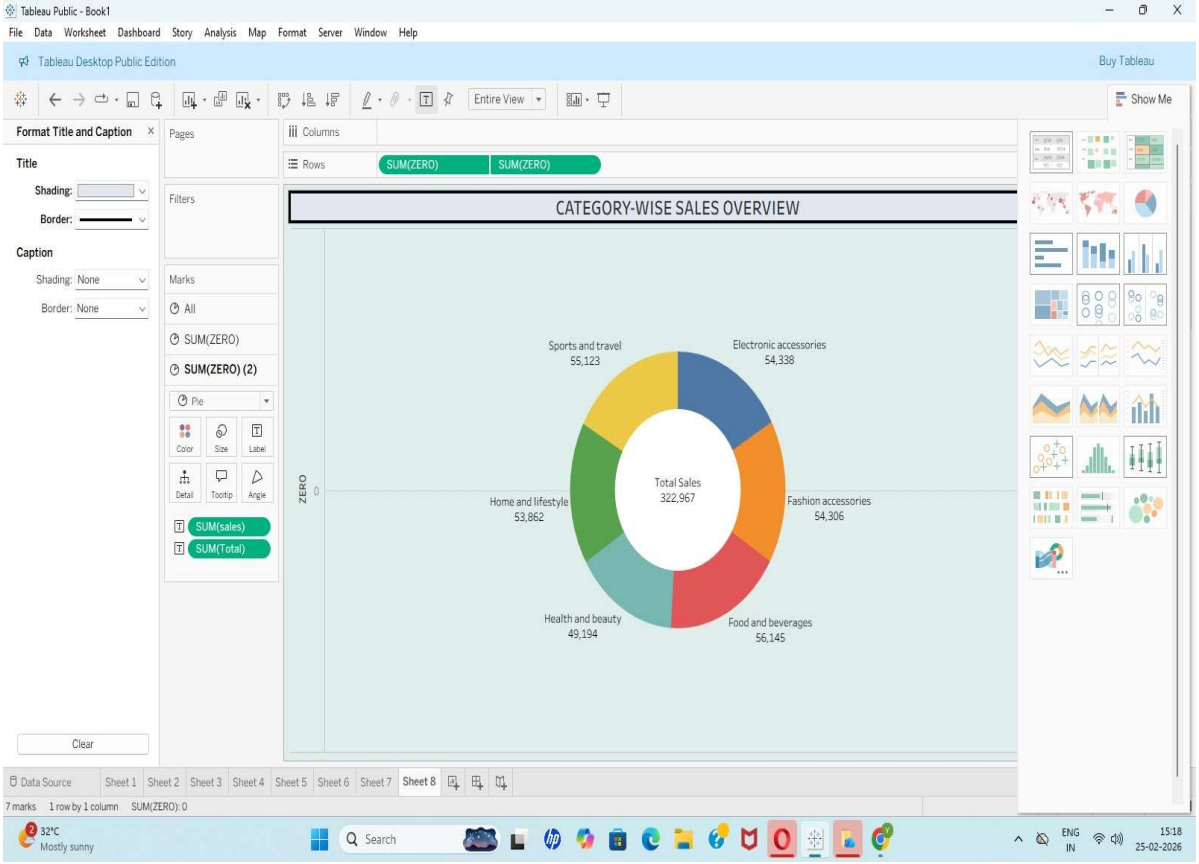
1. Drag Total Sales (SUM(Sales)) to Label of inner pie
2. Format text → Align center
3. Edit label to show:
   Total Sales
   322,967

Step 5: Add Title & Formatting

1. Click Worksheet → Show Title
2. Edit title to:
   CATEGORY-WISE SALES OVERVIEW
3. Format background color and borders (optional)

OUTPUT:

DONUT CHART: CATEGORY-WISE SALES OVERVIEW



2.Prepare Area Chart:

PURPOSE:

- Monitor monthly sales changes **(January, February, March)**
- Understand each branch's contribution **to total sales**

STEPS:

Step– 01: Create a New Sheet and Renamed as "Area Chart"

Step–02: Drag Date →Columns Shelf-Click the drop-down on Date Select Month

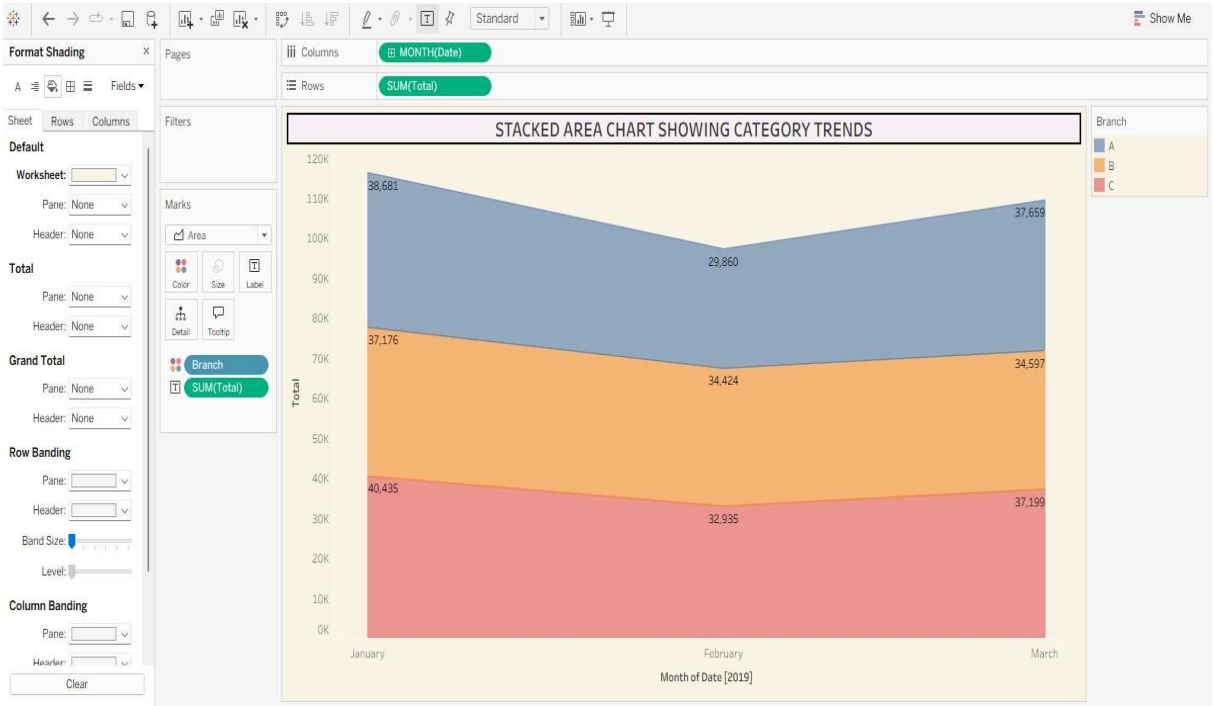
Step-03: Drag Total → Rows shelf

Step-04: Click Show Me (top right) Select Area Chart

Step-05: Drag city → Colour (in Marks card)

OUTPUT:

AREA CHART: STACKED AREA CHART SHOWING CATEGORY TRENS



3. **Prepare Text Table:**
   PURPOSE:
   - Analyse sales performance by product line
   - Compare total sales across different product categories

   STEPS:

Step 1: Connect to Dataset

- Open Tableau
- Click Connect → Select your dataset (Excel/CSV)

Step 2: Create the View

1. Drag Product Line → Rows
2. Drag Total (Sales) → Marks → Text
3. Ensure aggregation is SUM(Total)

Step 3: Format the Table

1. Click Format
2. Adjust:
   - Background shading
   - Font size
   - Alignment (Right-align numbers)
3. Add alternating row shading (for better readability)

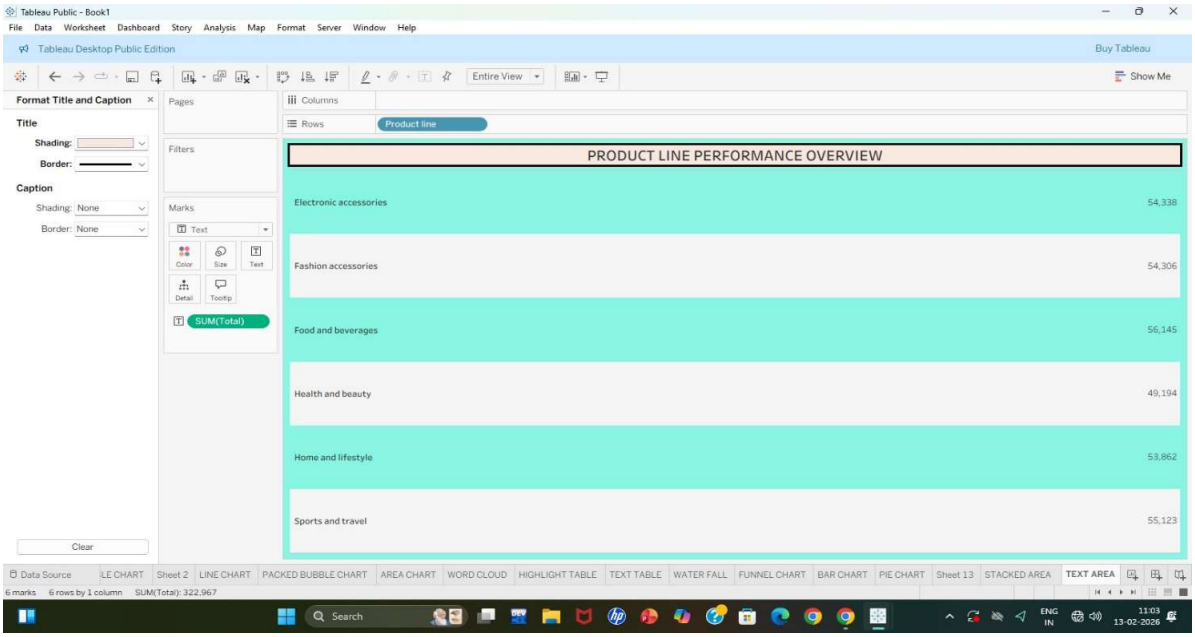
Step 4: Add Title

1. Click Worksheet → Show Title
2. Edit title to:
   PRODUCT LINE PERFORMANCE OVERVIEW
3. Format border and shading as needed

Step–05: Set Marks Type to Text-On Marks Card, click the dropdown-Select Text Table

OUTPUT:

TEXT TABLE: PRODUCT LINE PERFORMANCE OVERVIEW



4.Prepare Highlighted Table:

PURPOSE:

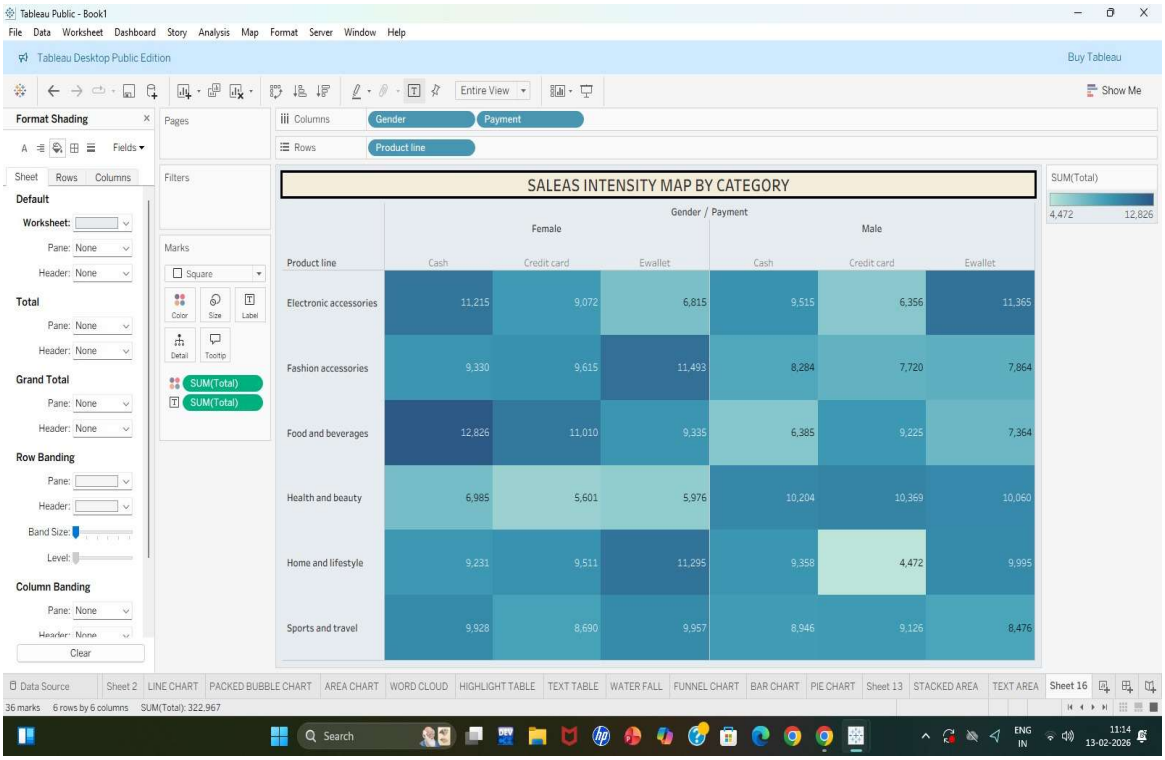
- Visualize sales performance using color intensity
- Quickly identify high and low sales categories

STEPS:

- Step–01: Create a New Sheet and Renamed as “Highlighted Table”
- Step–02: Drag Product line →Rows
- Step – 03: Drag Branch → Columns
- Step – 04: Drag Total → Marks → Colour on the Marks card, Click the drop-down → Select Square
- Step–05: In Marks card Size Drag slider to the right until squares fill the cells.
- Step – 06: Drag Total → Label -Click Label → Show Mark Labels
- Step–07: Click Colour Edit Colours Choose a colour palette Click Apply→Ok

OUTPUT:

HIGHLIGHTED TABLE: SALES INTENSITY MAP BY CATEGORY



6.Prepare Funnel Chart:

PURPOSE:

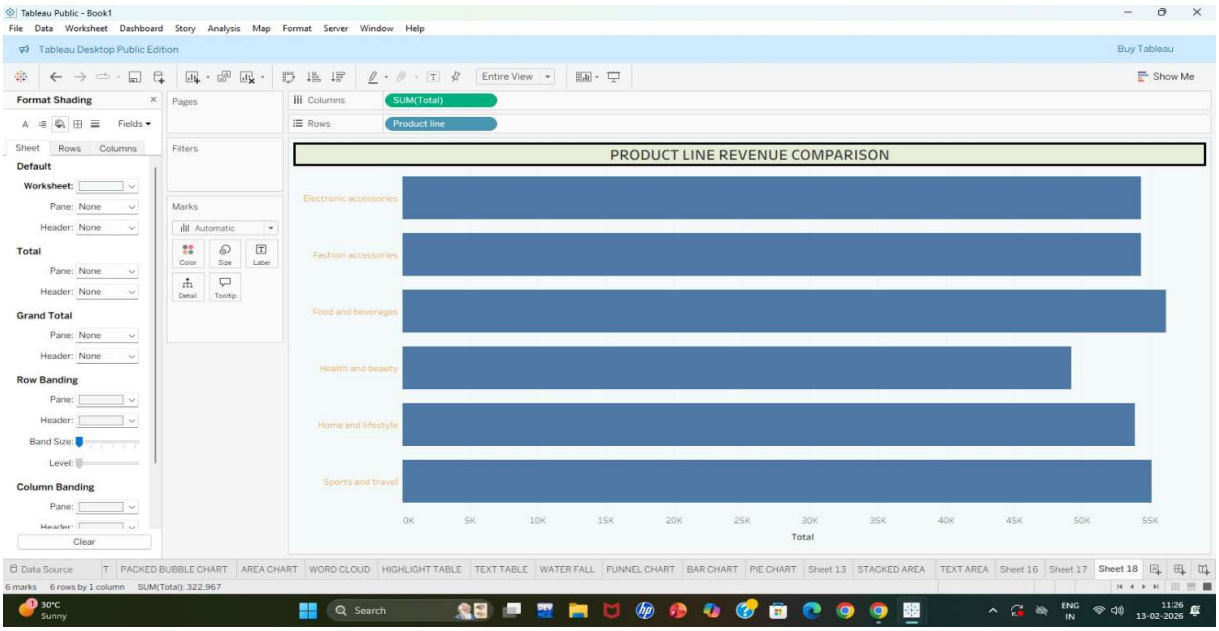
- Compare revenue across product lines
- Visualize ranking in descending order

STEPS:

Step–01: Create a New Sheet and Renamed as “Funnel Chart”
Step–02: On Marks Card-Select Bar
Step – 03: Drag Measure Values → Columns-Edit filters
Step – 04: Drag Measure Names→ Rows-It automatically shows in filter Click Sort Descending (largest to smallest)
Step–05: Right click on sheet-click on Format-Go to Format Shading-click on worksheet -change colour.

OUTPUT:

FUNNEL CHART: PRODUCT LINE REVENUE COMPARISON



7.PrepareWaterfall Chart

PURPOSE:

- Show profit generated by each product line
- Display cumulative (running) profit

STEPS:

Step–01: Create a New Sheet and Renamed as “Waterfall Chart”

Step–02: Drag Product line → Columns -Select All

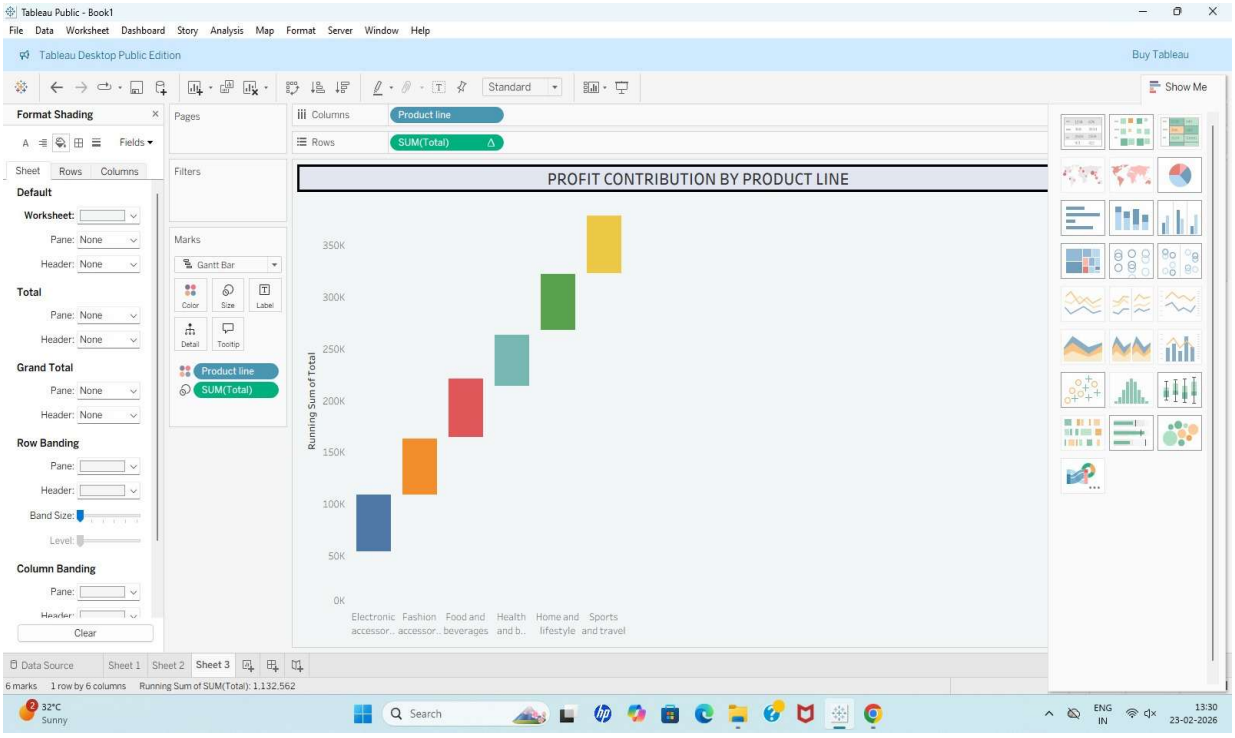
Step – 03: Drag Running Total→ Rows

Step–04: Click Show Me-Select Gantt Bar

Step–05: Right- click Create Calculated Field Name: Difference Formula: SUM([Total])- Click OK

Step – 06: Drag Difference → Size (Marks card)

Step–07: Drag Difference → Colour Waterfall Chart

<u>OUTPUT:</u>

WATERFALL CHART: PROFIT CONTRIBUTION BY PRODUCT LINE